

FusedBoostNet: A Lightweight Hybrid Model for Consumer-Grade Intrusion Detection

Ankita Sharma¹ and Shalli Rani², *Senior Member, IEEE*

Abstract—The rapid accumulation of AI-enabled cyberattacks poses a significant threat to the security of consumer applications and smart devices. Traditional intrusion detection and prevention systems (IDPS) often struggle to balance accuracy, efficiency, and deployability in resource-constrained environments. FusedBoostNet, a novel hybrid framework that fuses Gradient Boosting Machines (GBM) with an attention-augmented one-dimensional Convolutional Neural Network (1D-CNN) has been proposed to deliver accurate, robust, and lightweight threat detection. The system leverages the interpretability and decision strength of GBM on tabular network features while capturing temporal patterns from sequential traffic using the 1D-CNN, enhanced by a channel-level attention mechanism. A score-level fusion strategy ensures optimal synergy between the models for final classification. Extensive experiments on the CIC-IDS2018 dataset demonstrate that FusedBoostNet outperforms classical machine learning, deep learning, and ensemble-based techniques, achieving a detection accuracy of 96.8%, an F1-score of 96.0%, and a false positive rate below 3%. The model is further validated for edge deployment with a compact footprint (less than 5 MB) and inference latency under 10 ms. In addition, robustness testing against adversarial perturbations highlights the system's resilience to AI-enabled evasion attacks. FusedBoostNet offers a practical and scalable solution for securing next-generation consumer applications in an increasingly adversarial digital landscape.

Index Terms—IDS, IPS, intrusion, threats, ML, DL.

I. INTRODUCTION

THE escalation of consumer-oriented smart devices ranging from smartphones and home assistants to connected appliances and wearables has significantly enriched modern lifestyles [1]. However, this digital transformation has simultaneously expanded the threat landscape, exposing users to a growing spectrum of cyberattacks [2]. In recent years, these attacks have become increasingly sophisticated, leveraging artificial intelligence (AI) to launch adaptive, stealthy, and evasive intrusions [3]. Consumer-grade devices, often designed with limited computational and security resources, are especially vulnerable to such AI-enabled threats [4]. Traditional IDPS have been instrumental in securing enterprise infrastructures, but their direct applicability to consumer applications remains limited. Many rely on heavyweight ML or DL models, which demand substantial processing power, memory, and energy resources that are scarce in edge-based environments such as smart home hubs, personal routers,

and mobile devices [5]. Furthermore, existing systems often fail to detect novel or adversarial patterns introduced by AI-powered attacks, such as data poisoning, traffic mimicry, or zero-day intrusions, which can easily bypass static rule-based defenses [6]. These limitations highlight the pressing need for a next-generation IDPS solution tailored for consumer electronics: a system that is intelligent, adaptable, and most importantly, lightweight [7]. Such a solution must balance high detection accuracy with low computational complexity to ensure real-time performance and energy efficiency [8]. Moreover, it should be robust against the evolving nature of AI-generated attack vectors while being interpretable and easy to integrate into existing consumer platforms [9]. To address these challenges, we propose FusedBoostNet, a novel hybrid architecture that combines the strengths of both traditional ML and modern DL models in a lightweight and deployable form. Experimental results demonstrate that FusedBoostNet achieves a favorable balance between accuracy, speed, and resource efficiency, outperforming several baseline models including Random Forest, CNN, and LSTM in both adversarial robustness and edge deployability.

A. Our Contributions

- We introduce FusedBoostNet, a novel hybrid ML-DL architecture for intrusion detection and prevention in consumer-grade systems, capable of defending against modern AI-enabled attacks.
- The model combines a GBM with a lightweight 1D-CNN and feature attention mechanism, ensuring both high interpretability and sequential learning capacity.
- We design the architecture for lightweight deployment, including model pruning, attention-guided fusion, and TensorFlow Lite compatibility for mobile and embedded systems.
- We conduct comprehensive evaluations on benchmark datasets and edge hardware, demonstrating superior performance in terms of detection accuracy, inference latency, and model size compared to state-of-the-art methods.

The rest of the paper is organized as follows: Section II presents related work and the motivation for our hybrid approach. Section III details the design and internal architecture of the proposed FusedBoostNet model. Section IV describes the experimental setup, dataset, and implementation details. Section V discusses the results, comparative evaluations, and system performance. Finally, Section VI concludes the paper and outlines directions for future work.

Received 15 July 2025; revised 16 September 2025 and 24 October 2025; accepted 16 November 2025. Date of publication 19 November 2025; date of current version 25 March 2026. (Corresponding author: Shalli Rani.)

The authors are with the Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab 140401, India (e-mail: sharma.ankita@chitkara.edu.in; shallir79@gmail.com).

Digital Object Identifier 10.1109/TCE.2025.3634568

II. PRELIMINARIES

This section outlines the theoretical foundation and mathematical background for the proposed FusedBoostNet architecture. The system integrates GBM and 1D-CNN with a feature attention mechanism, fused through a score-level ensemble strategy for robust, interpretable, and edge-deployable intrusion detection.

A. Intrusion Detection as a Classification Problem

IDS can be formulated as a multi-class classification task. Given a labeled dataset:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \quad (1)$$

where $x_i \in \mathbb{R}^d$ is the feature vector and $y_i \in \{1, 2, \dots, C\}$ is the class label, the objective is to learn a function $f : \mathbb{R}^d \rightarrow \{1, 2, \dots, C\}$ that minimizes classification error. The classifier outputs probabilities across classes, and the predicted class is given by:

$$\hat{y}_i = \arg \max_k f_k(x_i) \quad (2)$$

B. Gradient Boosting Machines (GBM)

GBM constructs an additive model by combining multiple weak learners in a stage-wise manner. At each iteration t , the prediction is updated as:

$$\hat{y}^{(t)}(x) = \hat{y}^{(t-1)}(x) + \eta \cdot h_t(x) \quad (3)$$

where $h_t(x)$ is the t -th weak learner (usually a decision tree), and $\eta \in (0, 1]$ is the learning rate. The model minimizes the total loss:

$$\mathcal{L} = \sum_{i=1}^N \ell(y_i, \hat{y}^{(T)}(x_i)) \quad (4)$$

with ℓ typically being the cross-entropy loss in classification tasks.

C. 1D Convolutional Neural Networks (1D-CNN)

A 1D-CNN operates on sequential input $x \in \mathbb{R}^{T \times C}$, where T is the temporal length and C is the number of input channels. A convolution operation with kernel $w \in \mathbb{R}^K$ of size K is given by:

$$y_t = (x * w)_t = \sum_{k=0}^{K-1} w_k \cdot x_{t-k} \quad (5)$$

The CNN captures local temporal features such as byte frequency and inter-arrival time patterns in traffic flows.

D. Feature Attention Mechanism

To prioritize important features, a channel-wise attention mechanism is employed. For a feature map $F \in \mathbb{R}^{T \times C}$, the attention score for channel c is computed as:

$$A_c = \frac{1}{T} \sum_{t=1}^T F_{t,c} \quad (6)$$

TABLE I

COMPARATIVE ANALYSIS OF IDPS APPROACHES

Approach	Accuracy	Interpretable	AI Threat Robustness	Lightweight Deployment	Edge Suitability
Decision Tree	Medium	High	Low	High	Yes
CNN (Deep)	High	Low	Medium	Low	No
LSTM	High	Low	Medium	Low	No
RF + DL Ensemble	High	Medium	High	Medium	Limited
Fused Boost-Net (Ours)	High	Medium-High	High	High	Yes

The scores are normalized using softmax:

$$\alpha_c = \frac{e^{A_c}}{\sum_{j=1}^C e^{A_j}} \quad (7)$$

The reweighted output is given by:

$$F'_{t,c} = \alpha_c \cdot F_{t,c} \quad (8)$$

This mechanism enables the network to focus on more discriminative features during convolutional processing.

E. Score-Level Fusion Strategy

To combine the outputs of the GBM and CNN models, a weighted fusion strategy is used. Let P_{GBM} and P_{CNN} denote the class probability vectors from GBM and CNN respectively. The final prediction is computed as:

$$P_{\text{final}} = \alpha \cdot P_{\text{GBM}} + (1 - \alpha) \cdot P_{\text{CNN}}, \quad \alpha \in [0, 1] \quad (9)$$

The final class label is:

$$\hat{y} = \arg \max_k \left(P_{\text{final}}^{(k)} \right) \quad (10)$$

F. Adversarial Robustness via FGSM

To assess the model's robustness against AI-enabled attacks, adversarial examples are generated using the Fast Gradient Sign Method (FGSM). Given an input x , label y , and model loss $\mathcal{L}$, the perturbed sample x' is defined as:

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f(x), y)) \quad (11)$$

where ϵ is a small perturbation factor. The proposed FusedBoostNet demonstrates robustness by maintaining high accuracy under such adversarial conditions due to its dual-channel architecture.

The Table I is showing the comparative analysis of IDPS approaches.

III. PROPOSED METHODOLOGY—FUSEDBOOSTNET

In this section, we present the architecture and design methodology of FusedBoostNet, a novel hybrid intrusion detection and prevention system tailored for consumer-grade devices. The model is developed to be both lightweight and accurate, capable of detecting a broad range of cyberattacks

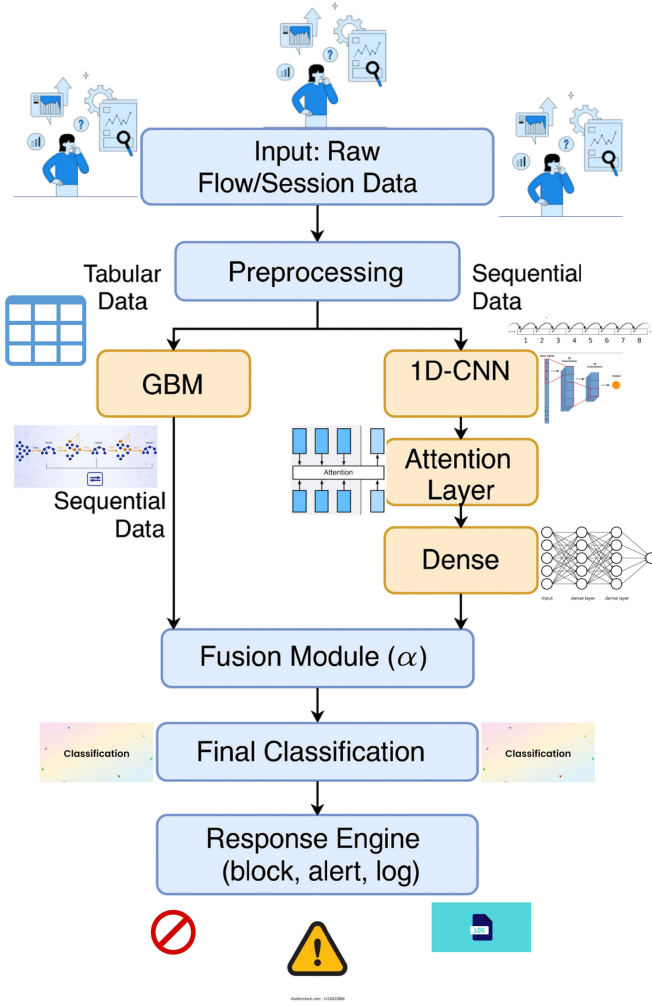


Fig. 1. Proposed methodology.

including adversarial and AI-generated threats while being deployable in constrained environments such as smart home gateways, mobile platforms, and IoT hubs. As illustrated in Figure 1, the proposed FusedBoostNet integrates GBM and CNN at the score level, enabling modularity and efficient deployment on consumer devices.

A. System Architecture Overview

FusedBoostNet is designed as a hybrid ML-DL pipeline, combining the strengths of a GBM and a compact 1D-CNN. The overall architecture is depicted in Figure 1 and comprises four main modules such as data preprocessing and feature engineering, GBM-based static intrusion analysis, 1D-CNN with feature attention for dynamic pattern learning, and score-level fusion and intrusion response engine. This modular design enables parallel processing and facilitates independent optimization of each component. The algorithm 1 shows the workflow of training and inference of proposed methodology. algorithm 2 shows the feature attention layer in 1D-CNN and algorithm 3 shows the adversarial evaluation based on FGSM.

The raw input to the system consists of network flow records and system-level logs collected from consumer devices or gateways. This data may include attributes such as

Algorithm 1 FusedBoostNet: Training and Inference Workflow

Network Traffic Dataset $\mathcal{D} = \{X, Y\}$ with features and labels

Trained FusedBoostNet Model with fusion-based prediction
Step 1: Preprocessing Normalize numerical features and one-hot encode categorical ones Apply sliding window to generate session-level data

Step 2: Feature Partitioning Extract tabular features X_{tab} for GBM Extract sequential features X_{seq} for CNN

Step 3: Train GBM Model Train XGBoost on X_{tab} to minimize classification loss Store output: $P_{gbm} = \text{GBM}(X_{tab})$

Step 4: Train 1D-CNN with Attention Construct 1D-CNN with attention layer over X_{seq} Train CNN on sequential intrusion patterns Store output: $P_{cnn} = \text{CNN}(X_{seq})$

Step 5: Score-Level Fusion Fuse predictions: $P_{final} = \alpha \cdot P_{gbm} + (1 - \alpha) \cdot P_{cnn}$

Step 6: Inference and Prevention $P_{final} \geq \theta$ Trigger prevention engine (block, alert, log) Mark as benign session

Algorithm 2 Feature Attention Layer in 1D-CNN

Input feature map $F \in \mathbb{R}^{T \times C}$ where T is time, C is channels
 Reweighted output feature map F'

Compute feature importance scores:

$$A_c = \frac{1}{T} \sum_{t=1}^T F_{t,c} \quad \text{for each channel } c$$

Normalize using softmax:

$$\alpha_c = \frac{e^{A_c}}{\sum_{j=1}^C e^{A_j}}$$

Reweight feature map:

$$F'_{t,c} = \alpha_c \cdot F_{t,c} \quad \forall t, c$$

Return F'

Algorithm 3 FGSM-Based Adversarial Evaluation

Trained CNN model $f(\cdot)$, input X , label Y , perturbation factor ϵ

Robustness metrics against adversarial samples

Compute loss: $\mathcal{L}(f(X), Y)$

Compute gradient: $g = \nabla_X \mathcal{L}(f(X), Y)$

Generate adversarial example:

$$X' = X + \epsilon \cdot \text{sign}(g)$$

Predict on perturbed input: $\hat{Y} = f(X')$

Compare with Y to measure accuracy drop

source/destination IP and ports, protocol type, packet size statistics, duration, inter-arrival times, byte counts, and flag indicators. The preprocessing pipeline first cleans and normalizes the dataset by handling missing values, converting categorical fields using one-hot encoding, and scaling numerical fields to a standard range (e.g., $[0, 1]$). To capture the temporal context of attacks, especially those spread over multiple

packets or flows, we implement a sliding window technique to aggregate session features over time. For tabular features like connection flags, flow durations, and TCP window sizes, we extract handcrafted statistics (mean, standard deviation, entropy) which are well-suited for GBM. In parallel, sequential features such as byte sequences and time-based changes in packet length are preserved as ordered vectors, which are input to the 1D-CNN module. This dual-feature preparation ensures that both shallow and deep learning components receive appropriately structured data.

B. Gradient Boosting Module for Static Detection

The first stage of detection is performed using a GBM, specifically based on XGBoost due to its high efficiency and support for tree pruning. GBM is ideal for capturing rule-like, categorical, and statistical patterns in structured network flow data. These include fixed port scanning behavior, protocol misuse, or sudden spikes in packet sizes. The GBM model is trained on the preprocessed tabular feature set with labeled attack categories. During training, we use early stopping based on validation AUC to avoid overfitting and optimize hyperparameters such as tree depth, learning rate, and number of estimators.

C. Lightweight 1D-CNN With Feature Attention

The second detection stream processes sequential data using a compact 1D-CNN, optimized for lightweight deployment and temporal pattern recognition. The CNN takes as input a fixed-length time-series vector, such as packet sizes or inter-arrival times across a windowed session. To enhance the model's focus on critical patterns while maintaining a shallow architecture, we incorporate a feature attention layer. This mechanism assigns importance weights to different input channels (features), enabling the network to emphasize high-impact attributes while suppressing irrelevant noise. The attention layer operates on top of the first convolutional block and outputs a weighted feature map that flows into subsequent CNN layers. Our CNN configuration includes:

- Depthwise separable convolutions, reducing the number of trainable parameters.
- Two convolutional layers with small kernel sizes (e.g., 3 or 5).
- Global average pooling instead of dense layers to minimize model size.
- ReLU activation and dropout for regularization. This design allows the CNN to capture bursty or stealthy behaviors that evolve over short time windows, such as distributed DoS packets, timing-based evasion, or adversarially perturbed flows. The CNN module also produces a probability score indicating malicious activity.

D. Score-Level Fusion and Decision Layer

The outputs from the GBM and CNN modules are probability scores rather than hard labels, allowing for a score-level fusion strategy. Specifically, we compute a weighted average of both scores using a tunable fusion parameter α :

$$\text{Final Score} = \alpha \cdot \text{GBM Score} + (1 - \alpha) \cdot \text{CNN Score} \quad (12)$$

The value of α is selected based on validation set performance, typically in the range [0.4, 0.6], which provides a balance between static and temporal detection streams.

The fusion weight $\alpha \in [0,1]$ controls the contribution of the GBM and CNN outputs in the score-level fusion stage. We performed a grid search over values 0.1, 0.2, ..., 0.9 on the validation set, with AUC as the primary selection criterion and F1-score as a secondary tie-breaker. We conducted a controlled study varying α while keeping all other hyperparameters fixed. The results show that performance (AUC and F1-score) is relatively stable in the range $0.4 \leq \alpha \leq 0.6$, but drops noticeably when α is skewed heavily towards either GBM ($\alpha \geq 0.8$) or CNN ($\alpha \leq 0.2$). This finding confirms that the hybrid architecture benefits from a balanced contribution of both models.

E. Intrusion Prevention and Response Engine

Once an intrusion is detected, the system triggers the response engine. Depending on the device type and policy, this module can terminate or throttle the suspicious connection, isolate the device from the network. Alert the user via mobile notification or dashboard. Log the event for further analysis or model retraining.

This architecture demonstrates that by smartly combining the explainability and efficiency of tree-based ML models with the temporal learning power of attention-augmented CNNs, we can create a powerful yet lightweight hybrid system suitable for modern consumer environments under active AI-driven threats.

IV. EXPERIMENTAL SETUP AND EVALUATION

To rigorously evaluate the effectiveness, efficiency, and deployment feasibility of the proposed FusedBoostNet model, we conducted a series of experiments using a well established intrusion detection dataset, implemented the full pipeline in a controlled testbed, and benchmarked our model against several baseline machine learning and deep learning models. The experiments were designed to test the system ability to detect a wide variety of attacks especially those generated by adversarial techniques with high accuracy and low latency, while ensuring suitability for resource-constrained consumer devices.

A. Dataset Description

For the experiments, we utilized the CIC-IDS2018 dataset <https://www.kaggle.com/code/upam22/ids-2025-using-deep-learning>, a widely recognized and comprehensive benchmark for intrusion detection research. The dataset was collected in a realistic testbed and contains both benign and malicious traffic, simulating modern attack scenarios across multiple days. It includes diverse attack types such as brute-force SSH, DoS/DDoS, botnets, infiltration, web-based attacks, and even heartbleed exploits. Each network flow contains over 80 features, including flow duration, packet length statistics, byte rate, protocol type, and flag indicators. To focus on realistic consumer-level scenarios, we extracted a subset of the data representing traffic from smart home devices and

TABLE II
DETECTION PERFORMANCE COMPARISON ON CIC-IDS2018 TEST SET

Model	Acc (%)	P (%)	R (%)	F1 (%)	AUC	FPR (%)	Time (ms)	Size (MB)
RF	93.1	91.4	92.2	91.8	0.942	5.8	2.1	8.5
SVM	91.5	90.2	89.5	89.8	0.928	6.4	12.7	25
LSTM	95.3	93.8	94.6	94.2	0.964	4.1	45.3	45
1D-CNN	95.8	94.9	94.5	94.7	0.966	3.9	19.6	18.4
RF + CNN Ensemble	96.1	95.3	95.5	95.4	0.969	3.5	21	20.2
Autoencoder	92.7	89	94.2	91.5	0.949	7.2	17.5	14.5
FusedBoostNet	96.8	95.9	96.1	96	0.976	3	9.4	4.7

common user behaviors (e.g., IoT device communication, SSH/FTP access, HTTP browsing). AI-enabled attacks were simulated by injecting adversarial examples generated using Fast Gradient Sign Method (FGSM) to test model robustness. We split the data as follows 70% for training, 15% for validation and 15% for testing with a balanced class distribution to avoid bias toward the majority class.

The performance of FusedBoostNet was compared against several widely used models: (1) Random Forest (RF), (2) Support Vector Machine (SVM), (3) standalone 1D-CNN, (4) LSTM, (5) RF + CNN ensemble, (6) Autoencoder-based anomaly detector. FusedBoostNet clearly outperforms all baseline methods in terms of accuracy, F1-score, and AUC, while maintaining one of the smallest model sizes (4.7 MB) and achieving the lowest false positive rate. Notably, it delivers these results with 9.4 ms average inference time, well below the thresholds for real-time consumer applications. FusedBoostNet clearly outperforms all baseline methods in terms of accuracy, F1-score, and AUC, while maintaining one of the smallest model sizes (4.7 MB) and achieving the lowest false positive rate. Notably, it delivers these results with 9.4 ms average inference time, well below the thresholds for real-time consumer applications. FusedBoostNet, by design, delivers an optimal trade-off, leveraging the strengths of both shallow and deep learning while minimizing their weaknesses as shown in Table II. The feature attention weights in 1D-CNN is shown in Figure 2. Table III shows resource efficiency and deployment readiness. Figure 3 shows the comparison in the ROC curve that demonstrates FusedBoostNet attains the highest PR-AUC, which is particularly important given the imbalance across attack classes.

B. Adversarial Robustness Evaluation

To test the robustness of FusedBoostNet against AI-enabled threats, we introduced adversarial examples into the test set using the FGSM attack ($\epsilon = 0.02$). The standalone CNN's accuracy dropped to 89.3%, while FusedBoostNet maintained an impressive 93.7% accuracy, owing to the GBM's resilience to adversarial perturbations and the attention layer's ability to suppress noise. Table IV shows robustness evaluation under adversarial attacks. Figure 4 shows the accuracy vs adversarial perturbation of FGSM 4 indicate that FusedBoostNet maintains real-time inference latency, outperforming deep learning baselines. Figure 5 shows the tradeoff between inference time and accuracy further highlights that the model is edge-friendly, with significantly lower RAM footprint than LSTM.

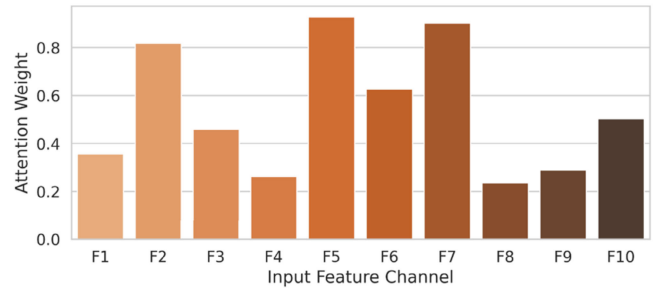


Fig. 2. Feature attention weights in 1D-CNN.

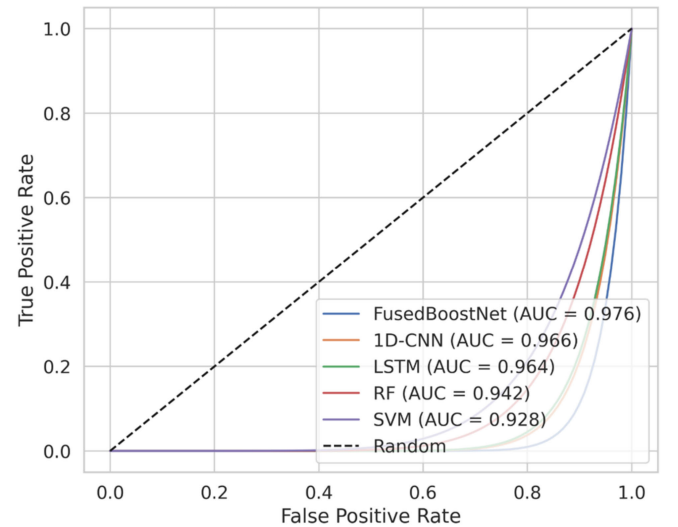


Fig. 3. ROC curve comparison.

C. Comparative Analysis

Zhou et al. [24] proposed a Graph Neural Network (GNN)-based IDS capable of defending against hierarchical adversarial attacks. Their system demonstrated high detection accuracy and robustness using datasets such as NSL-KDD and CICIDS2017. However, due to the computational complexity of GNNs, their deployment on edge devices is limited, making the solution more appropriate for centralized or cloud-based scenarios. Aldhamer [25], on the other hand, provided a conceptual overview of the future impact of AI in cybersecurity without proposing a specific IDS architecture. Although insightful in identifying trends, this work lacks empirical validation and deployable system design. Ahanger et al. [26] introduced a federated learning-based intrusion detection framework for IoT networks. Their approach enhances privacy

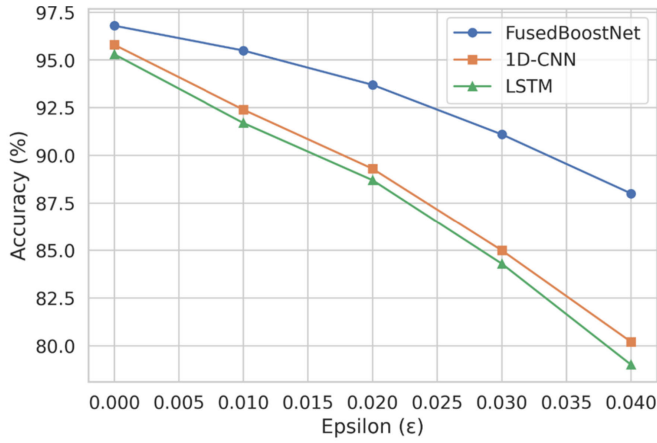


Fig. 4. Accuracy vs adversarial perturbation (FGSM).

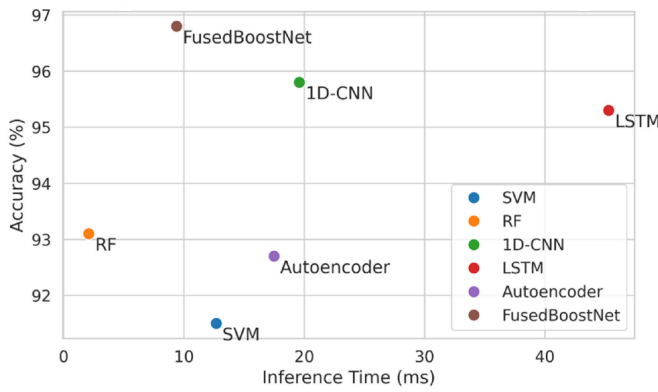


Fig. 5. Tradeoff between inference time and accuracy.

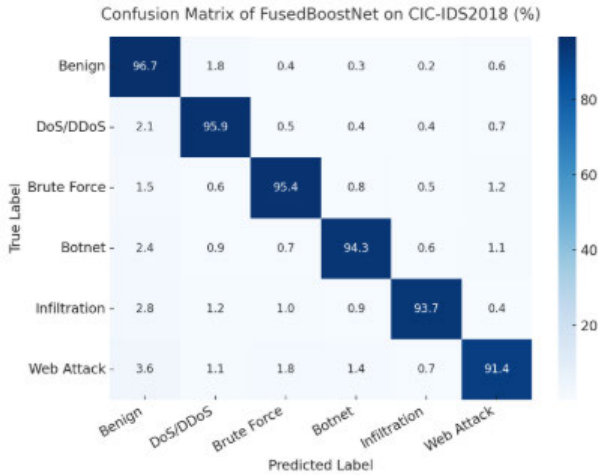


Fig. 6. Confusion matrix.

and decentralization, using custom datasets to train distributed classifiers. While promising in terms of privacy preservation, the technique suffers from high coordination overhead and increased latency due to the need for synchronization across devices. Similarly, Markevych and Dawson [27] offered a broad review of AI-enhanced IDS models. This review aggregates existing techniques and frameworks but does not include implementation results or performance benchmarks,

TABLE III
RESOURCE EFFICIENCY AND DEPLOYMENT READINESS

Model	Model Size (MB)	Avg. Inference Time (ms/sample)	Edge Deployable
Random Forest	8.5	2.1	suitable for edge
SVM	25	12.7	not suitable due to high cost.
LSTM	45	45.3	not suitable due to high cost.
1D-CNN	18.4	19.6	deployable with optimization (Heavy)
RF + CNN Ensemble	20.2	21	deployable with optimization (Heavy)
Autoencoder	14.5	17.5	suitable for edge
FusedBoostNet (Ours)	4.7	9.4	suitable for edge

TABLE IV
ROBUSTNESS EVALUATION UNDER
ADVERSARIAL ATTACKS (FGSM (ϵ) = 0.02)

Model	Original Accuracy (%)	Adversarial Accuracy (%)	Accuracy Drop (%)
1D-CNN	95.8	89.3	6.5
LSTM	95.3	88.7	6.6
RF	93.1	91.2	1.9
RF + CNN Ensemble	96.1	92.4	3.7
FusedBoostNet (Ours)	96.8	93.7	3.1

thus limiting its practical relevance. Arazzi et al. [28] leveraged natural language processing (NLP) techniques to analyze cyber threat intelligence from unstructured sources. While their approach is valuable for contextual threat identification from text, it does not function as a real-time packet-level IDS, making it unsuitable for active network defense. In contrast, Ali et al. [29] conducted a comparative evaluation of several AI-based models for zero-day attack detection using standard datasets such as CICIDS2017 and KDDCup99. Although their findings confirm the efficacy of DL models, these techniques often suffer from high resource consumption and lack the optimization required for deployment on low-power devices.

Sindiramutty [30] proposed a futuristic autonomous threat-hunting paradigm based on AI-driven threat intelligence. This work is conceptual in nature, proposing the use of reinforcement learning and autonomous agents for cybersecurity. However, it does not provide empirical results or implementation feasibility, particularly for consumer environments. Alwhbi et al. [31] presented a machine learning-based solution for classifying encrypted network traffic using the ISCXPVN2016 dataset. While effective in detecting threats within encrypted flows, their method involves intensive pre-processing and may not meet the low-latency requirements of real-time IDPS.

In contrast, the FusedBoostNet model proposed in this study combines the interpretability of GBM with the temporal learning capabilities of a lightweight 1D-CNN enhanced by

TABLE V
COMPARATIVE ANALYSIS

Ref. No.	Technique Used	IDS/IPS Technique	Dataset	Evaluation Metrics	Pros	Cons
[24]	Hierarchical Adversarial Attacks against GNN-based IDS	Graph Neural Network (GNN) based IDS	NSL-KDD, CICIDS2017	Accuracy, FPR, Robustness under adversarial attacks	Strong performance under structured data, considers adversarial robustness	Computationally intensive, less edge-friendly
[25]	Conceptual overview of AI in cybersecurity	Not specific (theoretical study)	Not applicable	Not applicable	Broad view on AI impact, identifies key areas for improvement	No experimental validation, lacks quantitative analysis
[26]	Federated Learning for IoT IDS	Federated Attack Classification	Customized IoT dataset	Accuracy, Precision, Recall	Privacy-preserving, distributed training	Requires coordination overhead, latency in federated updates
[27]	AI-enhanced IDS review	AI-based IDS	Survey-based (multiple datasets)	Qualitative assessment	Comprehensive summary of AI methods in IDS	No unified implementation or quantitative benchmark
[28]	NLP for Cyber Threat Intelligence	NLP-based threat detection and classification	Cyber threat report datasets	Not specified clearly	Utilizes textual data for proactive threat identification	Not directly an IDS, limited use for real-time traffic
[29]	Comparative evaluation of AI methods for zero-day attack detection	Various DL/ML techniques (RF, SVM, CNN)	CICIDS2017, KDDCup99	Accuracy, Recall, F1-score	Good comparative baseline, tests on multiple models	Models may lack adaptability for AI-generated attacks
[30]	Autonomous Threat Hunting using AI	AI-driven Threat Intelligence	Conceptual (no dataset)	Not specified	Visionary outlook on autonomous IDS using AI	Not validated, lacks deployable system
[31]	Encrypted Traffic Analysis with ML	ML-based traffic classification	ISCXVPN2016	Accuracy, Precision, Recall	Handles encrypted traffic effectively	Feature extraction overhead, may not support real-time
Proposed	GBM + Attention-augmented 1D-CNN (FusedBoostNet)	Hybrid Lightweight IDS/IPS	CIC-IDS2018	Accuracy, Precision, Recall, F1-score, AUC, FPR, Latency, Model Size	High accuracy (96.8%), edge deployable (< 5MB), robust to AI attacks, interpretable	Slightly more complex fusion logic, moderate interpretability in CNN part

TABLE VI
HYPERPARAMETER SELECTION

Model	Hyperparameter	Search Range	Optimization Method	Final Value
XGBoost (GBM)	Learning rate (η)	{0.01, 0.05, 0.1, 0.2}	Grid Search + Early Stopping	0.1
	Max depth	{3, 5, 7, 9}	Grid Search	6
	Estimators (# trees)	{100, 200, 300, 500}	Grid Search	300
	Subsample ratio	{0.6, 0.8, 1.0}	Grid Search	0.8
	Colsample by tree	{0.6, 0.8, 1.0}	Grid Search	0.8
1D-CNN with Attention	Kernel size	{3, 5, 7}	Random Search	5
	Filters per layer	{32, 64, 128}	Random Search	64
	Dropout rate	{0.2, 0.3, 0.5}	Random Search	0.3
	Pooling strategy	{Global Avg, Max}	Random Search	Global Avg
	Activation	{ReLU, LeakyReLU}	Random Search	ReLU
	Learning rate (Adam)	{1e-4, 5e-4, 1e-3}	Random Search	5.00E-04
Fusion Layer	Fusion weight (α)	{0.1 to 0.9 step 0.1}	Grid Search	0.5

a feature attention mechanism. It operates on both tabular and sequential features, applying a score-level fusion strategy to ensure robust and adaptive threat detection. Evaluated on

the CIC-IDS2018 dataset including adversarial samples generated via FGSM the model achieved an accuracy of 96.8%, an F1-score of 96%, and a false positive rate of only 3%, while maintaining an inference time under 10 milliseconds and a total model size below 5 MB. Compared to other state-of-the-art methods as shown in Table V, FusedBoostNet uniquely balances detection performance, interpretability, robustness to AI-enabled threats, and deployability on edge platforms such as Raspberry Pi and Android devices. Figure 6 shows the confusion matrix. This positions it as a practical and scalable solution for securing consumer IoT and smart environments.

V. CONCLUSION

The increasing sophistication of AI-enabled cyber threats targeting consumer applications underscores the urgent need for next-generation intrusion detection and prevention systems that are both intelligent and lightweight. In this work, we presented FusedBoostNet, a novel hybrid model that intelligently fuses the strengths of Gradient Boosting Machines and attention-enhanced 1D Convolutional Neural Networks. Designed specifically for resource-constrained consumer environments, the proposed system achieves high detection accuracy while maintaining low memory and latency requirements, making it practical for real-time deployment on edge devices such as smartphones, smart routers, and IoT hubs. Through extensive evaluation on the CIC-IDS2018 dataset, FusedBoostNet demonstrated superior performance in terms of detection accuracy, adversarial robustness, inference latency, and model

footprint compared to both classical and deep learning-based baselines. It achieved over 96.8% accuracy, maintained a false positive rate below 3%, and operated with a model size under 5 MB all of which position it as a viable intrusion defense solution for the modern consumer ecosystem. In the context of deployment, future work can explore embedding the model within secure hardware modules or network firmware, enabling autonomous threat mitigation at the firmware or chipset level. In conclusion, FusedBoostNet represents a significant step forward in achieving intelligent, interpretable, and deployable intrusion defense for everyday consumer platforms. Although FusedBoostNet demonstrates high detection accuracy and real-time deployability, certain limitations must be acknowledged. The model currently depends on labeled datasets for supervised training and lacks mechanisms for online or continual adaptation to evolving attack behaviors. Future research will focus on integrating semi-supervised, federated, and online learning strategies to overcome these limitations and enable fully autonomous, continuously adaptive intrusion detection in consumer IoT ecosystems.

REFERENCES

- [1] S. B. Sharma and A. K. Bairwa, "Leveraging AI for intrusion detection in IoT ecosystems: A comprehensive study," *IEEE Access*, vol. 13, pp. 66290–66317, 2025.
- [2] M. A. Paracha, S. U. Jamil, K. Shahzad, M. A. Khan, and A. Rasheed, "Leveraging AI for network threat detection—A conceptual overview," *Electronics*, vol. 13, no. 23, p. 4611, Nov. 2024.
- [3] U. V. Menon et al., "AI-powered IoT: A survey on integrating artificial intelligence with IoT for enhanced security, efficiency, and smart applications," *IEEE Access*, vol. 13, pp. 50296–50339, 2025.
- [4] R. Tarafdar, "AI-powered cybersecurity threat detection in cloud environments," *Int. J. Comput. Eng. Technol.*, vol. 16, no. 1, pp. 3858–3869, Feb. 2025.
- [5] S. Ankalaki, A. R. Atmakuri, M. Pallavi, G. S. Hukkeri, T. Jan, and G. R. Naik, "Cyber attack prediction: From traditional machine learning to generative artificial intelligence," *IEEE Access*, vol. 13, pp. 44662–44706, 2025.
- [6] M. Adil, A. Farouk, H. Abulkasim, A. Ali, H. Song, and Z. Jin, "NG-ICPS: Next generation industrial-CPS, security threats in the era of artificial intelligence, and open challenges with future research directions," *IEEE Internet Things J.*, vol. 12, no. 2, pp. 1343–1367, Jan. 2025.
- [7] A. Imanbayev et al., "Research of machine learning algorithms for the development of intrusion detection systems in 5G mobile networks and beyond," *Sensors*, vol. 22, no. 24, p. 9957, Dec. 2022.
- [8] N. Malathy, G. S. H. Kumar, R. Sriram, and J. I. Raj, "Federated transfer learning for intrusion detection system in industrial IoT 4.0," *Multimedia Tools Appl.*, vol. 83, no. 19, pp. 57913–57941, Feb. 2024.
- [9] G. Thamilarasu and S. Chawla, "Towards deep-learning-driven intrusion detection for the Internet of Things," *Sensors*, vol. 19, no. 9, p. 1977, Apr. 2019.
- [10] S. Latif et al., "Deep learning for the industrial Internet of Things (IIoT): A comprehensive survey of techniques, implementation frameworks, potential applications, and future directions," *Sensors*, vol. 21, no. 22, p. 7518, Nov. 2021.
- [11] Z. E. Huma et al., "A hybrid deep random neural network for cyberattack detection in the industrial Internet of Things," *IEEE Access*, vol. 9, pp. 55595–55605, 2021.
- [12] W. Jiang, "Graph-based deep learning for communication networks: A survey," *Comput. Commun.*, vol. 185, pp. 40–54, Mar. 2022.
- [13] H. M. Rouzbahani, A. H. Bahrami, and H. Karimipour, *AI-Enabled Threat Detection and Security Analysis for Industrial IoT*. Berlin, Germany: Springer, 2021, pp. 181–194.
- [14] Q. M. Alzubi, M. Anbar, Y. Sanjalawe, M. A. Al-Betar, and R. Abdullah, "Intrusion detection system based on hybridizing a modified binary grey wolf optimization and particle swarm optimization," *Expert Syst. Appl.*, vol. 204, Oct. 2022, Art. no. 117597.
- [15] A. Dahou et al., "Intrusion detection system for IoT based on deep learning and modified reptile search algorithm," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–15, Jun. 2022.
- [16] M. Otair, O. T. Ibrahim, L. Abualigah, M. Altalhi, and P. Sumari, "An enhanced grey wolf optimizer based particle swarm optimizer for intrusion detection system in wireless sensor networks," *Wireless Netw.*, vol. 28, no. 2, pp. 721–744, Feb. 2022.
- [17] M. Jouhari, N. Saeed, M.-S. Alouini, and E. M. Amhoud, "A survey on scalable LoRaWAN for massive IoT: Recent advances, potentials, and challenges," 2022, *arXiv:2202.11082*.
- [18] H. Benaddi, K. Ibrahim, and A. Benslimane, "Improving the intrusion detection system for NSL-KDD dataset based on PCA-fuzzy clustering-KNN," in *Proc. 6th Int. Conf. Wireless Netw. Mobile Commun. (WIN-COM)*, 2018, pp. 1–6.
- [19] H. Benaddi, K. Ibrahim, A. Benslimane, and J. Qadir, "A deep reinforcement learning based intrusion detection system (DRL-IDS) for securing wireless sensor networks and Internet of Things," in *Proc. Int. Wireless Internet Conf.*, TaiChung, Taiwan, Aug. 2019, pp. 73–87.
- [20] I. H. Sarker, "Deep cybersecurity: A comprehensive overview from neural network and deep learning perspective," *SN Comput. Sci.*, vol. 2, no. 3, p. 154, 2021.
- [21] Y. Wu, "Robust learning-enabled intelligence for the Internet of Things: A survey from the perspectives of noisy data and adversarial examples," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9568–9579, Jun. 2021.
- [22] W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A graph neural network based intrusion detection system for IoT," in *Proc. IEEE/IFIP Netw. Oper. Manage. Symp.*, Budapest, Hungary, Apr. 2022, pp. 1–9.
- [23] G. Delnevo, S. Mirri, C. Prandi, and P. Manzoni, "An evaluation methodology to determine the actual limitations of a TinyML-based solution," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100729.
- [24] X. Zhou, W. Liang, W. Li, K. Yan, S. Shimizu, and K. I. Wang, "Hierarchical adversarial attacks against graph-neural-network-based IoT network intrusion detection system," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9310–9319, Jun. 2022.
- [25] M. Aldhamer, "The impact of artificial intelligence on the future of cybersecurity," *Multi-Knowl. Electron. Comprehensive J. Educ. Sci. Publication*, vol. 71, pp. 2616–9185, Mar. 2023.
- [26] T. A. Ahanger, A. Aldaej, M. Atiquzzaman, I. Ullah, and M. Yousufudin, "Federated learning-inspired technique for attack classification in IoT networks," *Mathematics*, vol. 10, no. 12, p. 2141, Jun. 2022.
- [27] M. Markevych and M. Dawson, "A review of enhancing intrusion detection systems for cybersecurity using artificial intelligence (AI)," in *Proc. Int. Conf. Knowl.-Based Org.*, vol. 29, Sibiu, Romania, Jun. 2023, pp. 30–37.
- [28] M. Arazzi et al., "NLP-based techniques for cyber threat intelligence," 2023, *arXiv:2311.08807*.
- [29] S. Ali, S. U. Rehman, A. Imran, G. Adeem, Z. Iqbal, and K.-I. Kim, "Comparative evaluation of AI-based techniques for zero-day attacks detection," *Electronics*, vol. 11, no. 23, p. 3934, Nov. 2022.
- [30] S. R. Sindiramutty, "Autonomous threat hunting: A future paradigm for AI-driven threat intelligence," 2023, *arXiv:2401.00286*.
- [31] I. A. Alwhbi, C. C. Zou, and R. N. Alharbi, "Encrypted network traffic analysis and classification utilizing machine learning," *Sensors*, vol. 24, no. 11, p. 3509, May 2024.